

The schematic diagram illustrates the RP2040-Zero board layout. The central component is the RP2040 chip (U1). It is connected to various components as follows:

- GP0** is connected to a **470R** resistor (R6) and a **47uF** capacitor (C5).
- GP1** is connected to a **47uF** capacitor (C5).
- GP29** is connected to a **10k** resistor (R4).
- GP28** is connected to a **10k** resistor (R5).
- GP27** is connected to a **200k** resistor (R1).
- GP26** is connected to a **100k** resistor (R2).
- GP25** is connected to a **10k** resistor (R3).
- GP24** is connected to a **10k** resistor (R3).
- GP23** is connected to a **10k** resistor (R3).
- GP22** is connected to a **10k** resistor (R3).
- GP21** is connected to a **10k** resistor (R3).
- GP20** is connected to a **10k** resistor (R3).
- GP19** is connected to a **10k** resistor (R3).
- GP18** is connected to a **10k** resistor (R3).
- GP17** is connected to a **10k** resistor (R3).
- GP16** is connected to a **10k** resistor (R3).
- GP15** is connected to a **10k** resistor (R3).
- GP14** is connected to a **10k** resistor (R3).
- GP13** is connected to a **10k** resistor (R3).
- GP12** is connected to a **10k** resistor (R3).
- GP11** is connected to a **10k** resistor (R3).
- GP10** is connected to a **10k** resistor (R3).
- GP9** is connected to a **10k** resistor (R3).
- GP8** is connected to a **10k** resistor (R3).
- GP7** is connected to a **10k** resistor (R3).
- GP6** is connected to a **10k** resistor (R3).
- GP5** is connected to a **10k** resistor (R3).
- GP4** is connected to a **10k** resistor (R3).
- GP3** is connected to a **10k** resistor (R3).
- GP2** is connected to a **10k** resistor (R3).
- GP1** is connected to a **10k** resistor (R3).
- GP0** is connected to a **10k** resistor (R3).

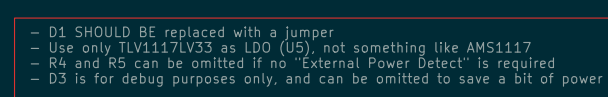
The board also includes a **3.3V** regulator (U2) and a **3.3V** regulator (U3). The board is labeled **RP2040-Zero**.

Diagram showing the wiring of the AHT10 temperature and humidity sensor to the Raspberry Pi 4:

- SDA (I2C data line) connected to Pin 4
- SCL (I2C clock line) connected to Pin 3
- VIN (5V supply) connected to Pin 1
- GND (Ground) connected to Pin 2

The sensor is powered by +3.3V and connected to GND.

The diagram illustrates a battery charging system. It includes a TP4056 Charge Controller (U4) and a CN3791 MPPT Solar Charger (U3). The TP4056 is connected to a Power In (J2) and a VBAT pin. The CN3791 is connected to a VIN pin and a BAT pin. A diode D1 (SS14) is connected between the BAT pin and the VBAT pin. The circuit is powered by a 5V regulator (U1) and a 3.3V regulator (U2).



Title: Zero Node